

14. (a) Sketch the graph of a function that has two local maxima, one local minimum, and no absolute minimum.
 (b) Sketch the graph of a function that has three local minima, two local maxima, and seven critical numbers.

15–28 Sketch the graph of f by hand and use your sketch to find the absolute and local maximum and minimum values of f . (Use the graphs and transformations of Sections 1.2 and 1.3.)

15. $f(x) = 3 - 2x, \quad x \geq -1$
 16. $f(x) = x^2, \quad -1 \leq x < 2$
 17. $f(x) = 1/x, \quad x \geq 1$
 18. $f(x) = 1/x, \quad 1 < x < 3$
 19. $f(x) = \sin x, \quad 0 \leq x < \pi/2$
 20. $f(x) = \sin x, \quad 0 < x \leq \pi/2$
 21. $f(x) = \sin x, \quad -\pi/2 \leq x \leq \pi/2$
 22. $f(t) = \cos t, \quad -3\pi/2 \leq t \leq 3\pi/2$
 23. $f(x) = \ln x, \quad 0 < x \leq 2$
 24. $f(x) = |x|$
 25. $f(x) = 1 - \sqrt{x}$
 26. $f(x) = e^x$

51–66 Find the absolute maximum and absolute minimum of f on the given interval.

51. $f(x) = 12 + 4x - x^2, \quad [0, 5]$
 52. $f(x) = 5 + 54x - 2x^3, \quad [0, 4]$
 53. $f(x) = 2x^3 - 3x^2 - 12x + 1, \quad [-2, 3]$
 54. $f(x) = x^3 - 6x^2 + 5, \quad [-3, 5]$
 55. $f(x) = 3x^4 - 4x^3 - 12x^2 + 1, \quad [-2, 3]$
 56. $f(t) = (t^2 - 4)^3, \quad [-2, 3]$
 57. $f(x) = x + \frac{1}{x}, \quad [0.2, 4]$
 58. $f(x) = \frac{x}{x^2 - x + 1}, \quad [0, 3]$
 59. $f(t) = t - \sqrt[3]{t}, \quad [-1, 4]$
 60. $f(x) = \frac{e^x}{1 + x^2}, \quad [0, 3]$
 61. $f(t) = 2 \cos t + \sin 2t, \quad [0, \pi/2]$
 62. $f(\theta) = 1 + \cos^2 \theta, \quad [\pi/4, \pi]$

29–48 Find the critical numbers of the function.

29. $f(x) = 3x^2 + x - 2$

30. $g(v) = v^3 - 12v + 4$

31. $f(x) = 3x^4 + 8x^3 - 48x^2$

32. $f(x) = 2x^3 + x^2 + 8x$

33. $g(t) = t^5 + 5t^3 + 50t$

34. $A(x) = |3 - 2x|$

35. $g(y) = \frac{y - 1}{y^2 - y + 1}$

36. $h(p) = \frac{p - 1}{p^2 + 4}$

37. $p(x) = \frac{x^2 + 2}{2x - 1}$

38. $q(t) = \frac{t^2 + 9}{t^2 - 9}$

39. $h(t) = t^{3/4} - 2t^{1/4}$

40. $g(x) = \sqrt[3]{4 - x^2}$

41. $F(x) = x^{4/5}(x - 4)^2$

42. $h(x) = x^{-1/3}(x - 2)$

43. $f(x) = x^{1/3}(4 - x)^{2/3}$

44. $f(\theta) = \theta + \sqrt{2} \cos \theta$